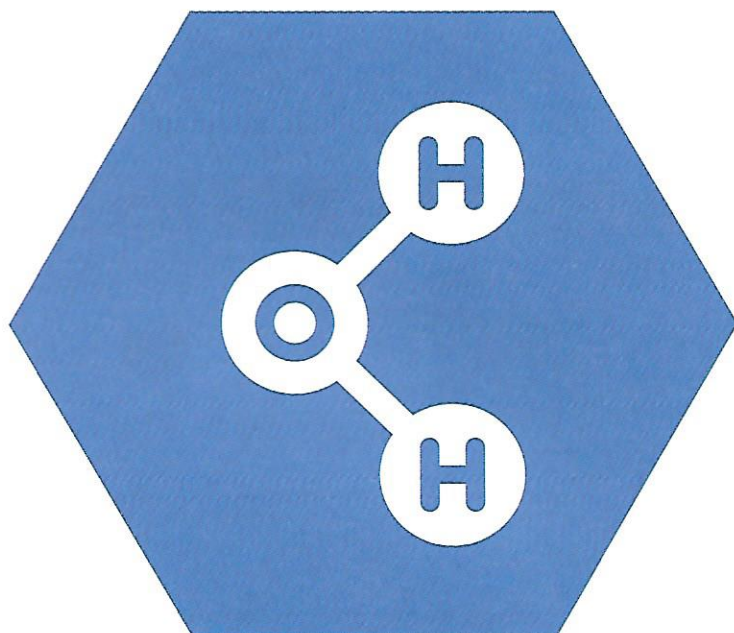




PAPER A



2016 ICAS

International Competitions
and Assessments for Schools

SCIENCE

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

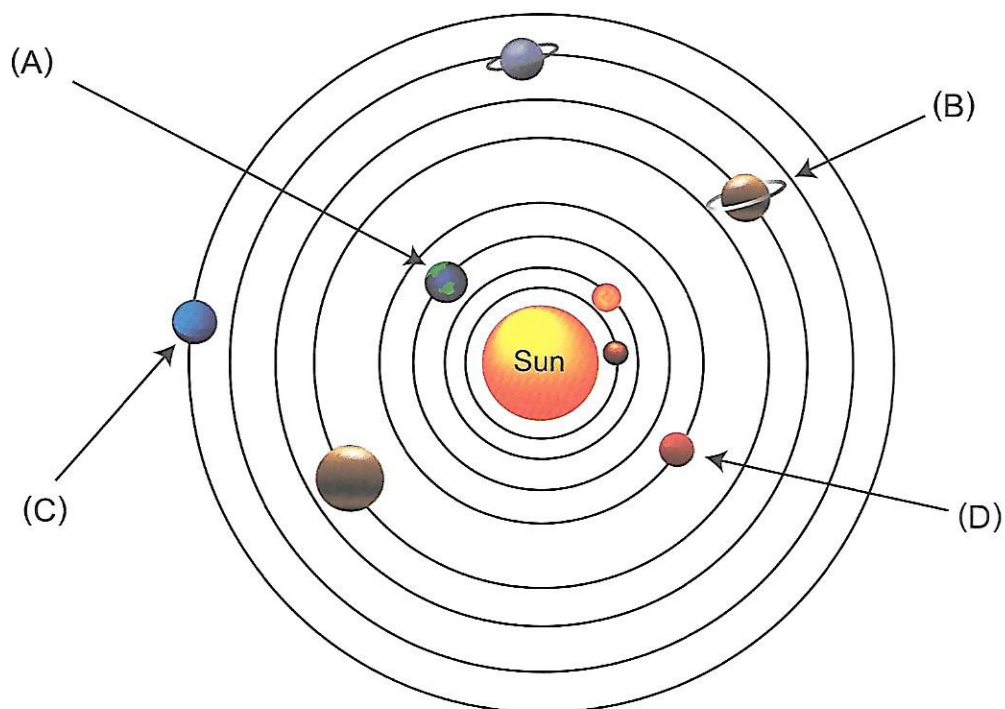
Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

1. The picture shows the planets in the Solar System as they move around the Sun in their orbits. Earth is the third planet from the Sun.

Which planet is Earth?



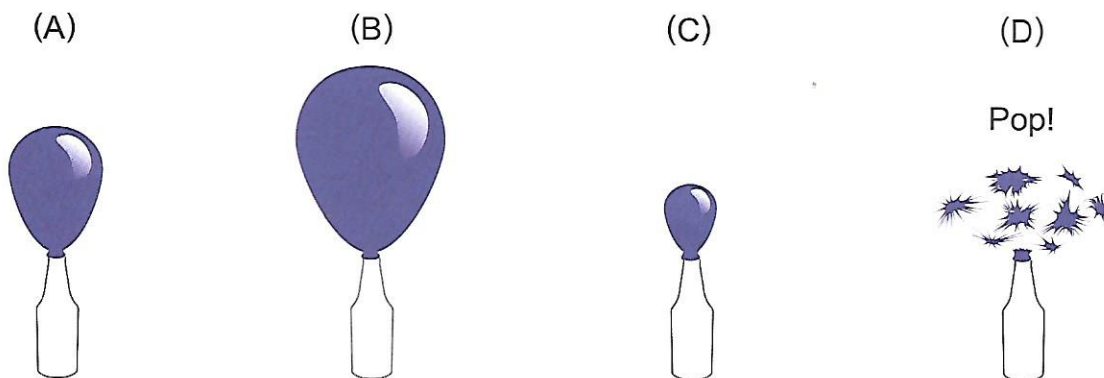
2. When air is cooled down, it is easier to squeeze it into a smaller space.

Mr Boot blew up a balloon and fitted it over the top of an empty bottle, as shown.



He then put the bottle in some icy water.

What did the balloon look like after the bottle had been in icy water?



3. When you make a metal hot enough, it will melt and turn into a liquid.

The table provides some information about the metal magnesium.

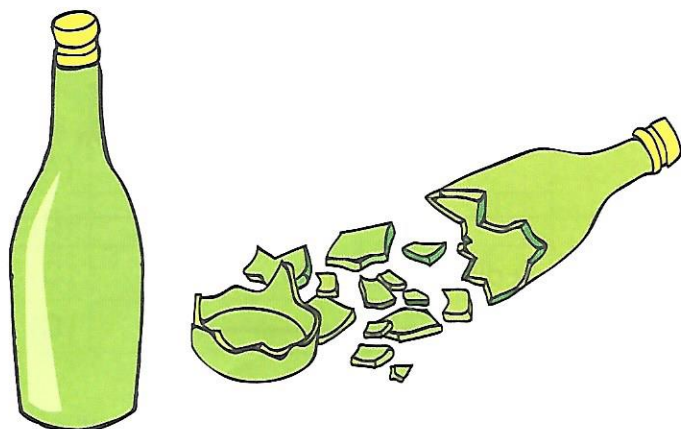
Information about magnesium	
Temperature at which it melts (°C)	850
Does it dissolve in water?	no
Does it let electricity go through it?	yes
Is it grey or black?	grey

What would a piece of magnesium look like at 25°C?

- (A) a grey solid
(B) a black solid
(C) a grey liquid
(D) a black liquid

4. Liam was carrying an empty glass bottle. He dropped the bottle and it broke.

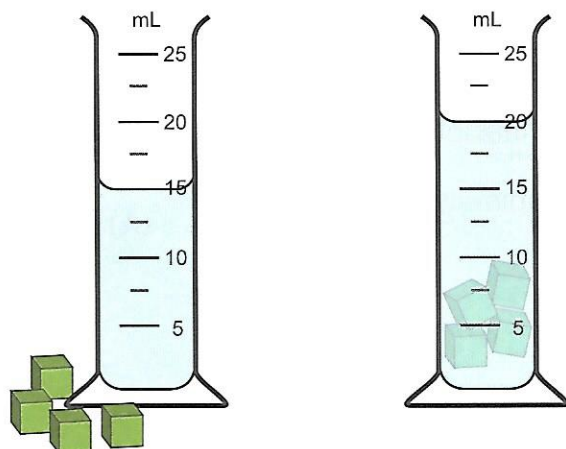
The drawing shows the bottle before and after Liam dropped it.



What is different about the glass bottle after he dropped it?

- (A) the amount of glass
(B) the shape of the bottle
(C) the colour of the bottle
(D) the material in the glass

5. When you place an object under water, it pushes the water out of the way and the water level goes up. This is called displacement.



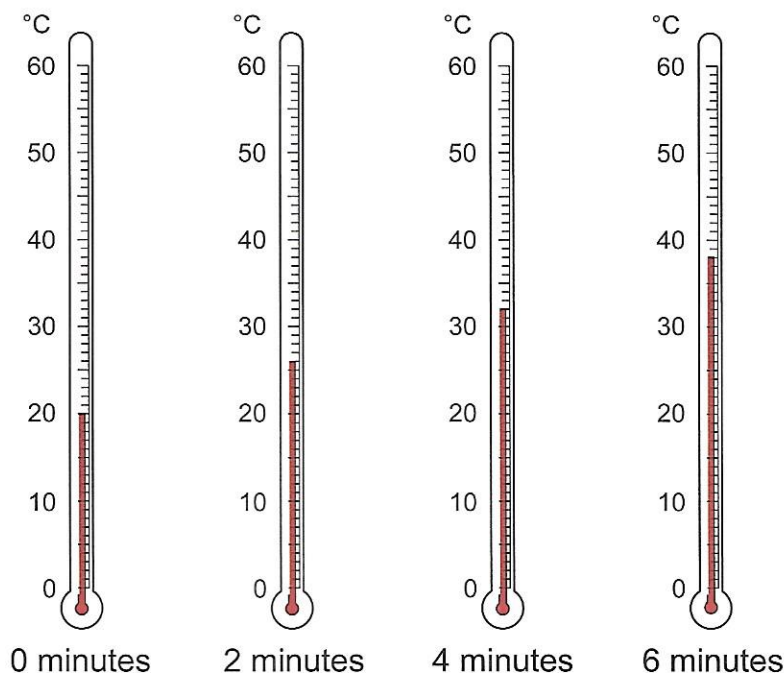
Jane placed 4 cubes in 15 millilitres (mL) of water:

How much water, in mL, did the cubes displace?

- (A) 4
(B) 5
(C) 15
(D) 20

6. Adam heated some water for eight minutes. He measured the water temperature using a thermometer.

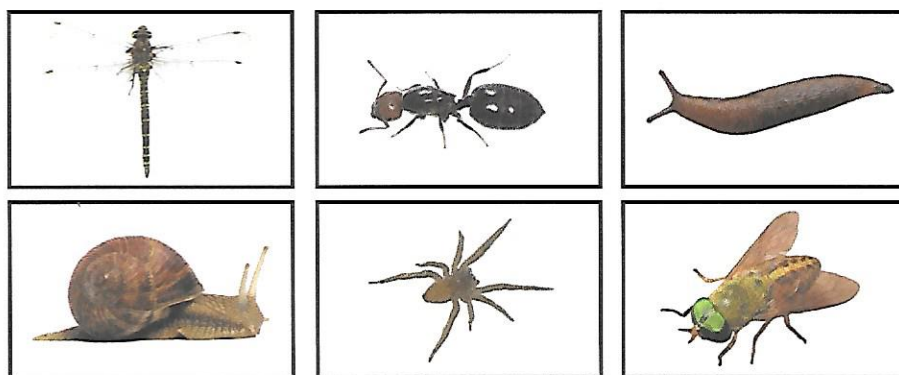
The thermometers show the temperature of the water every two minutes.



What will the temperature of the water be at 8 minutes?

- (A) 38 °C (B) 42 °C (C) 44 °C (D) 46 °C

7. Sunil's teacher gave him six photos, each showing a different animal.



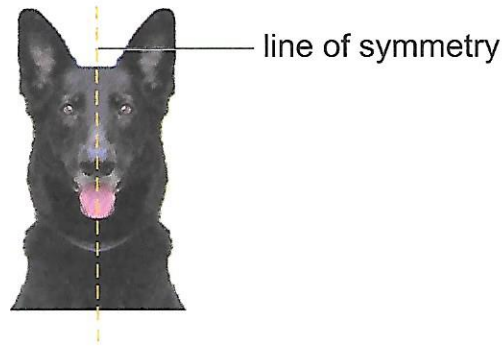
He told Sunil to think of a way to place the six animals into two groups.

Which of the following can he use to group these animals?

	Group 1	Group 2
(A)	with wings	without wings
(B)	with eyes	without eyes
(C)	big	small
(D)	old	young

8. Many living things are symmetrical.

For example, this picture of a dog has one line of symmetry.



The pictures below show a crab and a flower.



crab



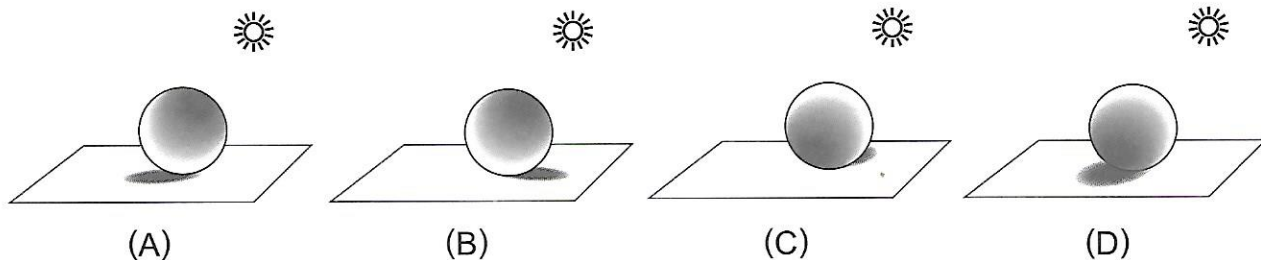
flower

Which option has the correct number of lines of symmetry for each picture?

	crab	flower
(A)	1	2
(B)	2	2
(C)	1	4
(D)	2	4

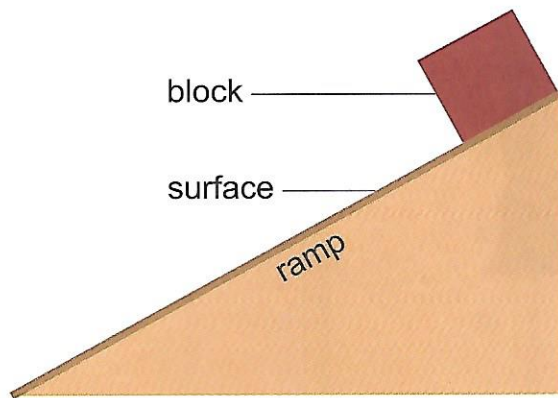
9. The sun is shining on a ball on a sheet of paper.

Which of these correctly shows the shadows on the ball and the paper?



10. Friction is a force that slows down the movement of objects.

Some students recorded the time it took for one block to slide down four ramps. The ramps were the same, except each ramp had a different surface.

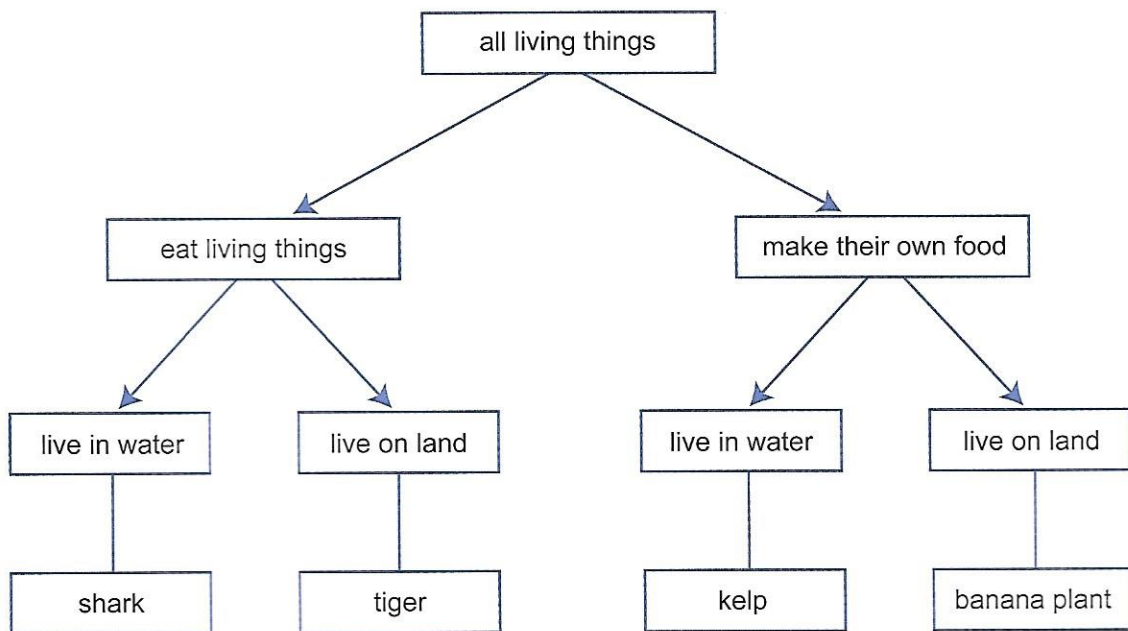


Surface	Time taken for block to slide down the ramp (seconds)
carpet	19
glass	5
wood	10
rubber	12

Which surface caused the most friction with the block?

- (A) carpet (B) glass (C) wood (D) rubber

11. The diagram shows one way in which living things can be placed into groups.

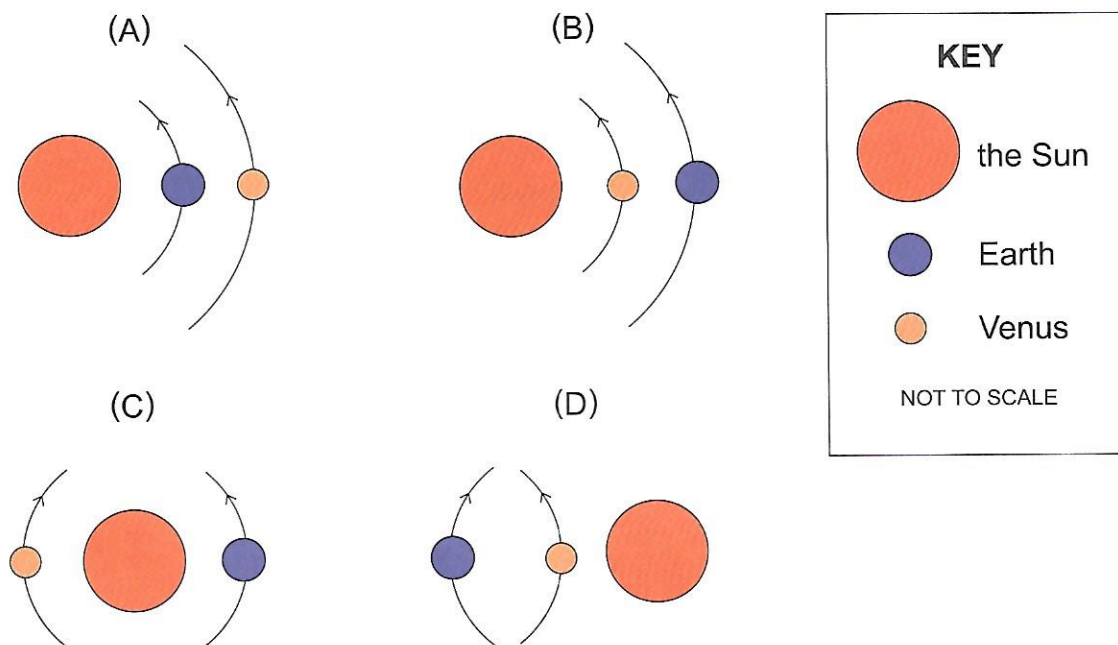


From the diagram, which of these living things makes its own food and lives in water?

- (A) shark (B) tiger
(C) kelp (D) banana plant

12. The transit of Venus occurs when the planet Venus travels between the Sun and Earth. It can be observed from Earth.

Which diagram correctly shows the position of the Sun, Venus and Earth when a transit of Venus takes place?



13. Sasi wanted to make some green slime.

These are the instructions she used:

- 1) In one bowl, mix $\frac{1}{2}$ cup glue and $\frac{1}{2}$ cup water. Add 3 drops food colouring.
- 2) In another bowl, mix 1 teaspoon borax with 1 cup water until the borax has dissolved.
- 3) Add the glue mixture to the borax solution, stirring slowly.

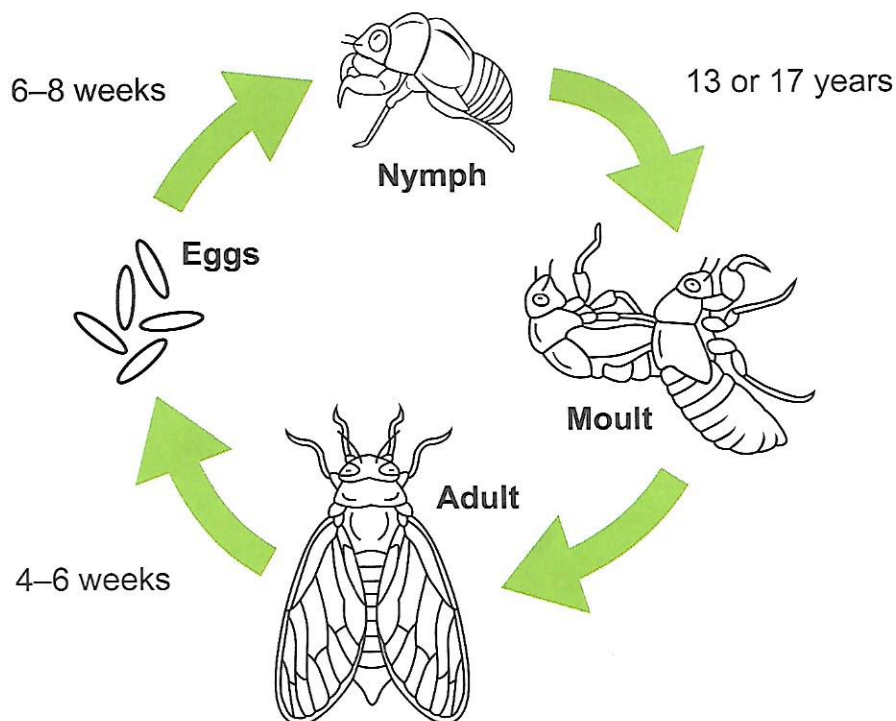
How many cups of water did Sasi need altogether?

- | | |
|--------------------|-------|
| (A) $\frac{1}{2}$ | (B) 1 |
| (C) $1\frac{1}{2}$ | (D) 2 |



For questions 14 and 15 use the information below.

Adult female cicadas in America lay eggs in the branches of trees. The eggs hatch into nymphs which then drop to the ground. The nymphs burrow underground and stay there for many years. Then they tunnel to the surface, crawl up the trees and shed their skin (moult). The adult female that comes out will mate and lay eggs.



14. How old is a cicada when it sheds its skin on a tree?

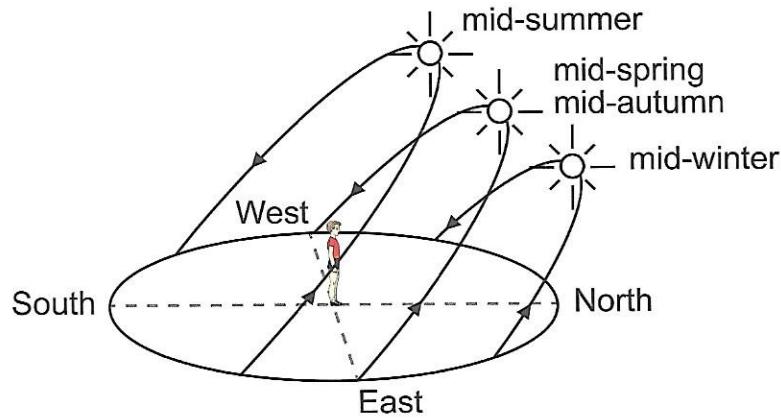
- | | |
|---------------|--------------------|
| (A) just born | (B) 4-6 weeks |
| (C) 6-8 weeks | (D) 13 or 17 years |

15. What body feature helps the nymph to reach the surface of the ground again?

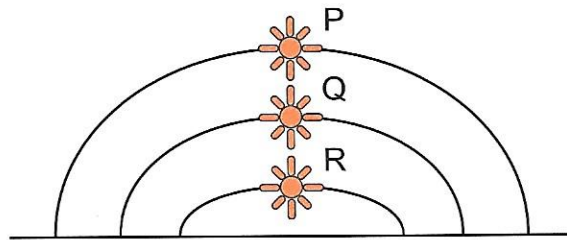
- (A) large wings
- (B) good eyesight
- (C) antennae on its head
- (D) strong front legs

16. This diagram shows the Sun's path across the sky in the southern hemisphere at different times of the year.

Jack stood on the same spot to look at the Sun at noon on days in mid-spring, mid-summer, mid-autumn and mid-winter.



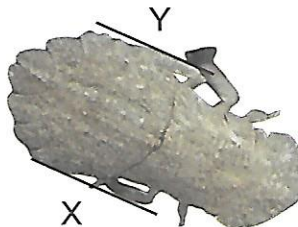
On each day, he drew the position of the Sun. He put them in one diagram.



Which of these options matches each position of the Sun with the time of year when he drew it?

	Summer	Autumn	Winter	Spring
(A)	P	Q	R	Q
(B)	P	Q	R	P
(C)	P	Q	Q	R
(D)	R	P	Q	P

17. Vivienne found a bug in her backyard. She measured the length of the bug. It was 1 centimetre long. This photo shows what the bug looks like.

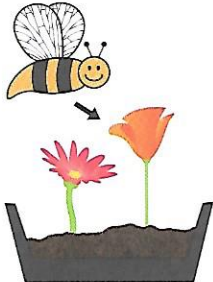
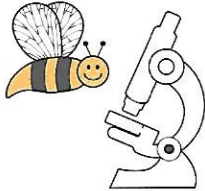
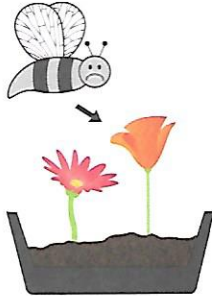


About how wide, in centimetres, is the bug from X to Y in real life?

- (A) 2 (B) $1\frac{1}{2}$ (C) 1 (D) $\frac{1}{2}$

18. Scientists want to test whether parasites can spread from infected bees to healthy bees when they collect pollen from the same flowers.

The drawings show four steps of the test. They are not in the correct order.

W	X	Y	Z
Healthy bees collect pollen from the flowers.	Bees are tested to see if they carry the parasites.	Bees carrying parasites are removed from the flowers.	Bees carrying parasites collect pollen from healthy flowers.
			

In which order do the steps have to be carried out to make this a correct test?

- (A) X, W, Z, Y (B) Z, Y, W, X (C) W, Y, X, Z (D) Y, X, Z, W

19. Cali wanted to find out what substances look like when they are mixed with water.

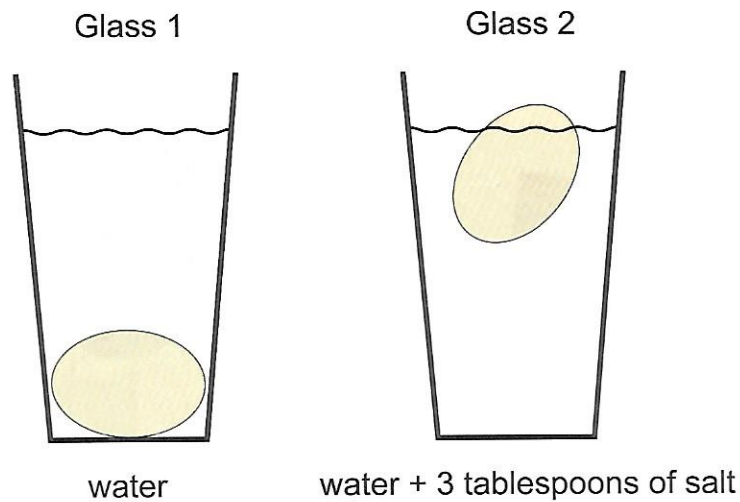
The table lists her results.

Substance mixed with water	What colour is the mixture?	Can you see through the mixture?
milk	white	no
mud	brown	no
salt	no colour	yes
sugar	no colour	yes
tea	brown	yes

After mixing with water, which two substances can Cali **NOT** tell apart using only the results of her investigation?

- (A) milk and mud (B) mud and tea
(C) salt and tea (D) salt and sugar

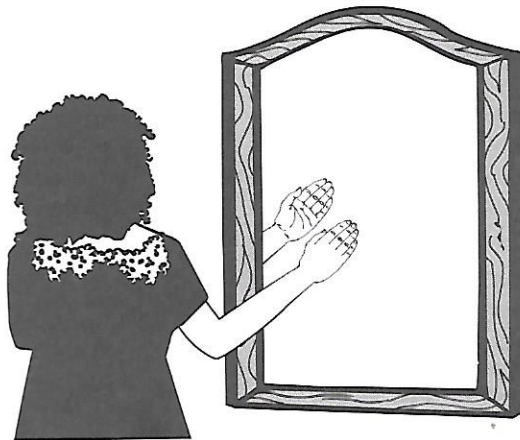
20. Mr Martin placed a fresh, raw egg into a glass of water. It sank to the bottom. He added 3 tablespoons of salt to a second glass of water and stirred it well. When he put the same egg into the salty water, it floated.



Boiled eggs float better than fresh eggs. Mr Martin boiled this raw egg and tested it in the same glasses.

Which of these statements must be true?

- (A) It floated in the salty water. (B) It sank in the fresh water.
(C) It floated in the fresh water. (D) It sank in the salty water.
21. When you see yourself in a mirror, you appear to be reversed. For example, when you hold up your right hand, its image in the mirror looks like your left hand.



Millie was wearing a dress with her name 'MILLIE' on the front. She stood in front of a mirror and looked at her name.

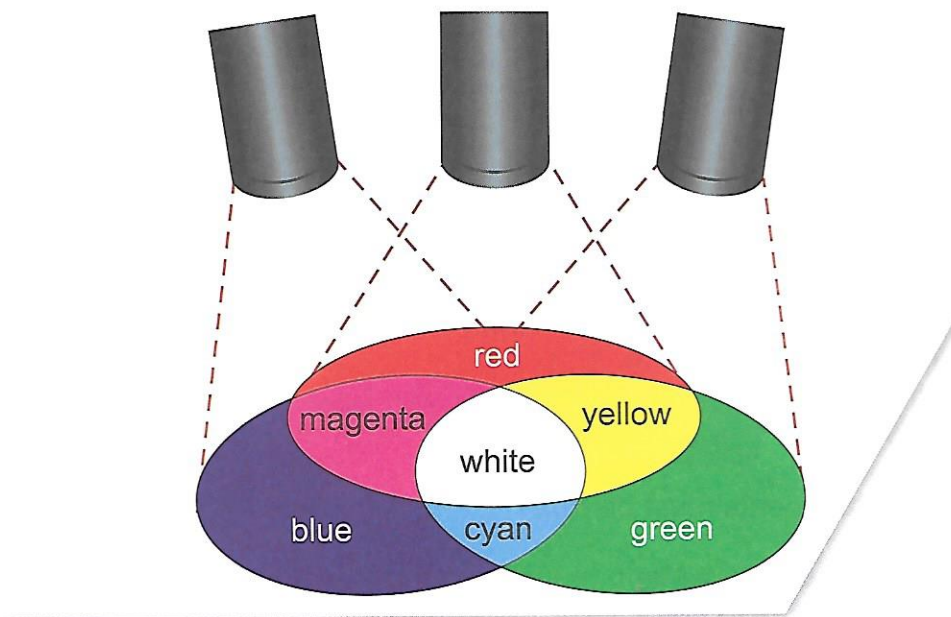
What did her name look like in the mirror?

- (A) MILLIE (B) EILLIM
(C) EILLIM (D) MILLIE

For questions 22 and 23 use the information below.

There are three colours of light called primary colours. When you mix two primary colours of light, you produce light with a secondary colour. Mixing all three primary colours in the right proportion produces white light.

Kate had three torches that each produced a different primary colour of light. She shone the torches onto a flat white surface, so their light beams overlapped.



22. Which of the following is a primary colour of light?
 (A) yellow (B) magenta (C) green (D) cyan
23. What colour of light do you get when you mix blue light and yellow light?
 (A) red (B) white (C) magenta (D) cyan

24. Tina has four different objects. She put them, two at a time, on a balance to see which object was heavier.

The diagrams show the results of her investigation.



Which option ranks the objects in order from lightest to heaviest?

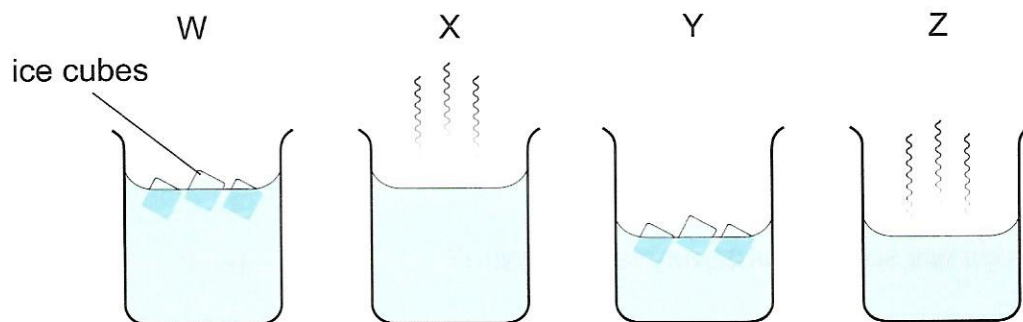
- (A) [White square] [Red circle] [Green triangle] [Purple hexagon]
 (B) [White square] [Green triangle] [Red circle] [Purple hexagon]
 (C) [Purple hexagon] [Green triangle] [White square] [Red circle]
 (D) [Purple hexagon] [Green triangle] [Red circle] [White square]

For questions 25 and 26 use the information below.

Adam knows that he can dissolve more sugar in hot water than he can in an equal amount of cold water.

He also knows that, at any temperature, the more water he has, the more sugar he can dissolve in it.

Adam has four cups of water.



25. Adam wants to dissolve as much sugar as possible.

Which cup of water must Adam use?

- (A) cup W (B) cup X
(C) cup Y (D) cup Z

26. He wants to order the cups according to how much sugar he can dissolve in each one.

Which two cups will be hardest to place in the correct order?

- (A) cup W and cup Z (B) cup X and cup Z
(C) cup W and cup Y (D) cup X and cup Y

27. The table shows the components that make up 100 grams of cow's milk.

Component	water	milk sugar	fat	protein	minerals
Amount (grams)	87	5	4	3	1

Only water is a liquid at room temperature. The rest of the components are solids.

Milk powder is made by removing the water from the milk.

How many grams of milk sugar is in milk powder made from 1000 grams of cow's milk?

- (A) 500 (B) 130 (C) 90 (D) 50

28. Seeds of different plants vary in the time they take to sprout at different temperatures.

The table compares the sprouting time of the seeds of four different plants.

Days it takes to sprout at different temperatures

	10 °C	15 °C	20 °C
onion seeds	13	7	5
spinach seeds	12	7	6
lettuce seeds	7	4	3
cabbage seeds	17	10	7

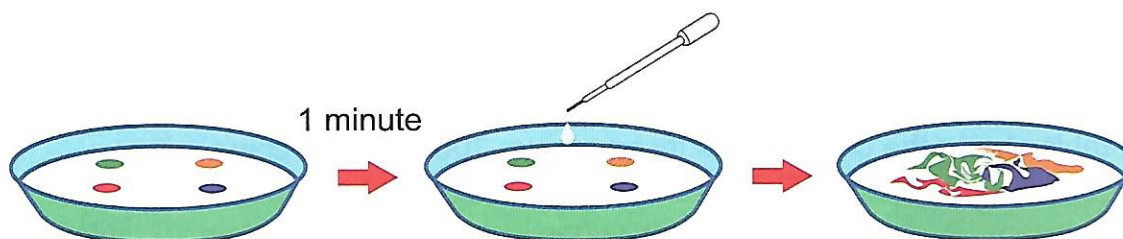
Which seeds will sprout first at any temperature?

- (A) onion
- (B) spinach
- (C) lettuce
- (D) cabbage

29. Mrs Black showed this science experiment to her Year 3 class.

She put a small drop of four food colours onto the surface of some milk in a dish.

After 1 minute she carefully added a drop of detergent onto the middle of the dish. The colours immediately started moving and mixing with each other forming colourful patterns on the surface of the milk.

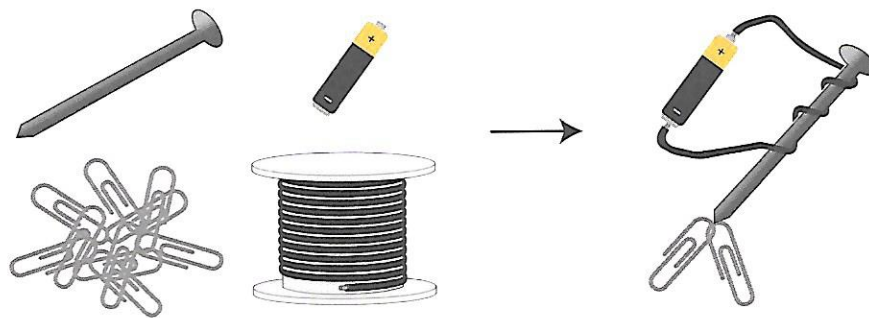


Which statement best explains these observations?

- (A) The milk surface is constantly moving, making the colours mix.
- (B) The milk surface is smooth, making the colours slide around.
- (C) The colours mix with the detergent, making the milk surface move and mix.
- (D) The detergent mixes with milk, making the milk surface move and the colours mix.

30. Xiao Ming read in a book that electricity can be used to turn a piece of iron into a magnet. He decided to test this idea.

He carried out his investigation using a large iron nail, some plastic-coated wire, a brand new battery and some paperclips. These drawings show what he did.



Which option correctly identifies whether this was a fair test and gives a correct reason?

	Is it a fair test?	What is the reason?
(A)	yes	The paperclips stuck to the nail with electricity around it.
(B)	yes	Electricity made the nail sticky.
(C)	no	He did not test if the nail alone would pick up the paperclips.
(D)	no	The battery is not powerful enough.

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Sources

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- Question 8** German Shepherd image adapted from a photograph by audreyjm529 <https://commons.wikimedia.org/wiki/File:German_Shepherd_Dog_black_sitting.jpg> licensed under the Creative Commons Attribution 2.0 Generic licence <<https://creativecommons.org/licenses/by-sa/2.0/>>.
- Question 13** Slime image by Nevit <https://commons.wikimedia.org/wiki/File:Slime_02471_Nevit.jpg> licensed under the Creative Commons Attribution-ShareAlike 3.0 Unported licence <<https://creativecommons.org/licenses/by-sa/3.0/>>.
- Questions 14–15** Life cycle diagram by Artsashina <<http://www.supercoloring.com/coloring-pages/life-cycle-of-a-cicada>> licensed under the Creative Commons Attribution-ShareAlike 4.0 International licence <<https://creativecommons.org/licenses/by-sa/4.0/>>.

The following year levels should sit THIS Paper:

Australia	Year 3
Brunei	Primary 3
Indonesia	Year 4
Malaysia	Standard 3
New Zealand	Year 4
Pacific Region	Year 4
Singapore	Primary 2
South Africa	Grade 3



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ANALYSIS BY QUESTION

The following table shows the questions you answered correctly ☒ and shades the questions that you answered incorrectly ☐. The questions are ranked from hardest at the top to easiest at the bottom.

	Question content	Area assessed	Question number	Correct answer
Difficult Questions	Identify the control in an investigation	Investigating	30	C
	Use reasoning to identify a difficulty in ordering a number of containers	Reasoning/Problem solving	26	A
	Estimate the width of a bug in a photo using a ratio	Observing/Measuring	17	D
	Order the steps of an investigation to make it a valid test	Investigating	18	B
	Calculate the mass of a portion using given data	Reasoning/Problem solving	27	D
	Interpret a diagram and given information related to coloured light	Interpreting	22	C
	Determine the result of mixing two colours of light	Reasoning/Problem solving	23	B
	Draw a conclusion based on the results of an investigation	Investigating	29	D
	Extrapolate the temperature reading from a scale	Observing/Measuring	6	C
	Determine the number of lines of symmetry for given objects	Observing/Measuring	8	C
	Use information provided to make a prediction	Predicting/Concluding	2	C
	Interpret information in a table	Interpreting	3	A
	Interpret information in a table	Interpreting	28	C
	Interpret diagrams relating the seasons and the position of the Sun in the sky	Predicting/Concluding	16	A
	Identify the mirror image of a word	Reasoning/Problem solving	21	B
	Rank objects by mass	Reasoning/Problem solving	24	A
	Draw a conclusion based on data in a table	Predicting/Concluding	19	D
	Use reasoning to determine the outcome of an action	Reasoning/Problem solving	20	A
	Use information provided to calculate the amount of water needed for a recipe	Investigating	13	C
	Interpret a diagram dealing with an astronomical phenomenon	Interpreting	12	B
Easy Questions	Identify a suitable criterion to categorise given animals into two groups	Interpreting	7	A
	Predict an outcome based on given information	Predicting/Concluding	25	B
	Identify where a shadow falls given the position of the light source	Observing/Measuring	9	D
	Interpret information in a diagram to determine the amount of liquid displaced by four cubes	Reasoning/Problem solving	5	B
	Predict the surface which provides the most friction	Predicting/Concluding	10	A
	Recognise a change in a property of an object	Predicting/Concluding	4	B
	Infer the function of a body feature of a cicada	Predicting/Concluding	15	D
	Interpret a flow chart to identify characteristics of some living things	Interpreting	11	C
	Interpret a diagram related to the life cycle of an organism	Interpreting	14	D
	Identify the position of Earth in a diagram of the Solar System	Interpreting	1	A